seq2seq_denoising.py

`src/moe_pruning/tasks/seq2seq_denoising.py · 87 lines · module moe_pruning.tasks.seq2seq_denoising`

What it does

The encoder-decoder objective used for Switch Transformers: `human` goes to the encoder, `response` is the decoder target; evaluation uses deterministic T5 span corruption.

How it works

- Loss = cross-entropy of the span targets (teacher forced); reported as a denoising loss, not perplexity.
- Qualitative output shows the original document, the corrupted input, the target and the model's greedy prediction.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class Seq2SeqDenoisingTask`	Encoder-decoder objective with T5 span-corruption evaluation.
`.tokenize_calibration(records, tokenizer, max_source_length, max_target_length)`	`human` is the input/prompt, `response` the teacher-forced target.
`.collate(items, pad_token_id)`	Right-padded batch with per-stack real-token masks.
`.eval_settings(cfg)`	Task-specific settings that determine the evaluation examples.
`.build_eval_example(example_id, text, tokenizer, cfg)`	One evaluation example (must expose `example_id`, `input_ids`, `labels`).
`.token_nll(logits, labels)`	Per-example (sum of token NLL, number of scored tokens).
`.predictions(logits, labels)`	Per-example argmax token at every scored position (same order as `token_nll`).
`.qualitative(model, tokenizer, examples, device, cfg)`	`[{"example_id", "inputs": {label: text}, "prediction"}]` from greedy generation.

Settings it reads

`EVAL_MEAN_SPAN_LENGTH`, `EVAL_NOISE_DENSITY`, `EVAL_SEED`, `MAX_SOURCE_LENGTH`, `MAX_TARGET_LENGTH`

Connections

Uses: `moe_pruning.data.calibration`, `moe_pruning.data.collators`, `moe_pruning.data.span_corruption`, `moe_pruning.tasks.base`

Used by: `src/moe_pruning/models/switch_transformers/__init__.py`